Figure 10.4. An example of absolute attention where a function might simply look at subregions of an image and only process those one at a time. This can significantly reduce the computational burden, since the dimensionality of each segment is much smaller than the whole image.

Figure 10.5. An example of soft attention where a model would learn which pixels to keep and which pixels to ignore (i.e., set to 0). Unlike the hard-attention model, the soft-attention model needs to process the entire image at once, which can be computationally demanding.

Figure 10.6. A recurrent neural network (RNN) is capable of maintaining an internal state that is updated with each new input it receives. This allows RNNs to model sequential data such as time series or language.

Figure 10.7. Schematic of an RNN language model. Two separate RNNs are used, an encoder and a decoder. The encoder takes an input sentence word by word and, once complete, sends its internal state to the decoder RNN, which produces each word in the target sentence until it halts.

Figure 10.8. Scatter plot of a nonlinear data set on which we might want to train a regression algorithm.

Figure 10.9. One way to perform nonparametric kernel regression to predict the y component of a new x value is to find the most similar (i.e., the closest) x's in the training data and then take the average of their respective y components.

Figure 10.10. The graph structure on the left can be represented quantitatively with a node feature matrix that encodes the individual node features and an adjacency matrix that encodes the edges (i.e., connections or arrows) between nodes. A 1 in the a row in the b column indicates that node a has an edge from a to b. The node features could be something like an RGBA value if the nodes represented pixels.

Figure 10.11. A relational module, at the highest level, processes a node matrix, , and outputs a new, updated node matrix, , where the dimensionality of the node feature may be different.

Figure 10.12. A convolutional layer returns a series of convolutional "filters" stored in a 3-tensor with shape channels (i.e., the number of